RUSHIL RAJESH

CFD / SCIENTIFIC COMPUTING / SOLVER DEVELOPER

CUDA • Numerical Methods • Poisson Solvers • LES • High-Performance Computing

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PROFILE

Mechanical Engineering M.S. researcher developing GPU-accelerated incompressible-flow solvers, with experience in numerical methods, Poisson projection, CUDA C++ backends, Python interfaces, benchmark validation, and performance optimization.

EDUCATION

University of Cincinnati | M.S., Mechanical Engineering — **May 2027**

GPA: 3.54/4.0

Vellore Institute of Technology | B.Tech, Mechanical Engineering — **May 2024**

GPA: 7.78/10

SOLVER DEVELOPMENT & RESEARCH

University of Cincinnati | Research Assistant - Dr. Shaban Abdallah — **Dec 2024 - Present**

M.S. Thesis - GPU-Accelerated Velocity-Impulse Incompressible-Flow Solver

- Design and implement incompressible velocity-impulse solvers with Runge-Kutta integration, finite-difference operators, scalar-potential projection, no-slip boundaries, and velocity recovery.
- Implement and benchmark iterative and transform-based Poisson strategies, including Jacobi/SOR-style relaxation, residual-based stopping, and direct DCT-based projection workflows to improve robustness and runtime.
- Develop Numba CUDA kernels in FP32 and optimize memory access, kernel launch structure, synchronization frequency, and residual checks for large 3D grids on NVIDIA GPUs.
- Validate against lid-driven-cavity benchmarks at Re = 100, 500, and 1000; track divergence norms, grid/time-step sensitivity, convergence behavior, and formulation-dependent numerical dissipation.

Independent Research — LES and Impulse-Based Turbulence Solver | Ongoing

- Implemented the simulation backend in CUDA C++, with a Python frontend and ctypes interface for run control, diagnostics, and post-processing.
- Developing a 3D geometric-impulse LES solver with WALE subgrid stresses; benchmark validation is in progress, alongside numerical-diffusion studies and CUDA kernel optimization.

SELECTED COMPUTATIONAL ENGINEERING PROJECTS

Aerospike Nozzle - Method of Characteristics + CFD — **May 2023 - Apr 2024**

- Modified a Python Method-of-Characteristics code to generate a truncated plug-nozzle contour, then connected the numerical geometry workflow to SolidWorks, ANSYS Fluent, thrust estimation, and prototype design.

Centrifugal Pump - Parametric CFD

- Created a repeatable CFD workflow for multiple impeller blade counts using TurboGrid and ANSYS Fluent, comparing pressure/head, loading, slip, blockage, and outlet-flow metrics.

TECHNICAL SKILLS

Programming & HPC: C++, CUDA C++, Python, ctypes, Numba, NumPy, MATLAB, GPU kernel optimization

Numerical Methods: Finite differences, collocated grids, projection methods, Poisson equations, DCT/FFT concepts, Jacobi/SOR, RK methods, residual analysis, grid/time-step studies

CFD & Turbulence: Incompressible flow, LES, WALE/Smagorinsky SGS, turbulence diagnostics, ANSYS Fluent, TurboGrid, heat transfer, turbomachinery